

Federated Learning Lab:

Environment Setup Instructions

Before starting the Federated Learning lab, please prepare your Python environment using the steps below.

Important: The laboratory notebooks have been tested using the following versions. Please use these versions to avoid compatibility problems.

Software / Package	Version
Python	3.11.7
Flower	1.16.0
Ray	2.31.0
Packaging	23.1

Step 1 - Open Anaconda Prompt

Windows users:

Open:

Start Menu → Anaconda Prompt

You may also use another terminal if the Conda command is available.

macOS → Terminal

Open:

Applications → Utilities → Terminal

Step 2 - Create the Lab Environment

Run the following command in the terminal:

```
conda create -n fl-summer python=3.11.7 -y
```

Then activate the environment:

```
conda activate fl-summer
```

After activation, the beginning of your terminal should look like:

```
(fl-summer) C:\Users\YourName> This indicates that the correct environment is active.
```

Step 3 - Install Flower and Ray

Run:

```
pip install flwr==1.16.0 ray==2.31.0 packaging==23.1
```

Wait until the installation is complete.

Step 4 - Install the Remaining Required Packages

Run:

```
pip install torch torchvision numpy matplotlib scikit-learn jupyter
```

These packages provide the tools required for neural network training, image datasets, visualization, evaluation, and Jupyter Notebook.

Step 5 - Start Jupyter Notebook

Make sure that the **fl-summer** environment is activated.

Then run:

```
jupyter notebook
```

Jupyter Notebook should automatically open in your web browser.

Navigate to the folder containing the laboratory notebooks and open the required notebook.

Step 6 - Verify Your Environment

In Jupyter Notebook, create a new cell and run:

```
import sys

import flwr

import packaging

import ray

import importlib.metadata

print("Python:", sys.version)

print("Flower:", importlib.metadata.version("flwr"))

print("Packaging:", packaging._version_)

print("Ray:", ray._version_)
```

The output should show approximately:

Python: 3.11.7

Flower: 1.16.0

Packaging: 23.1

Ray: 2.31.0

If these versions are displayed, your environment is ready.

Starting the Lab in Future Sessions

You only need to install the environment **once**.

The next time you want to work with the notebooks:

1. Open Anaconda Prompt
2. Activate the environment: **conda activate fl-summer**
3. Start Jupyter Notebook: **jupyter notebook**

Important Note: Always activate the **fl-summer** environment **before starting Jupyter Notebook**.

Otherwise, Jupyter may use another Python installation or environment, which can cause package and Flower compatibility errors.

Environment Ready

You are now ready to run the Federated Learning laboratory notebooks using:

PyTorch + Flower + Ray.